



# MATH230: Double Integrals

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Name: \_\_\_\_\_

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## Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

*Simplify each expression completely. Show all steps and circle your final answer.*

## Double integrals over rectangles

1. Evaluate the double integral: integral from 0 to 4 integral from 0 to 5 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [0, 5] \times [0, 4]$$

Answer: \_\_\_\_\_

2. A density function  $\rho = 2 \, \text{kg/m}^2$  is uniform over the rectangle  $[1, 3] \times [1, 4]$ . Find the total mass.

$$\iint_R 2 \, dA, [1, 3] \times [1, 4]$$

Answer: \_\_\_\_\_

3. Evaluate: integral from 0 to 2 integral from 0 to 5 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 5] \times [0, 2]$$

Answer: \_\_\_\_\_

4. Evaluate the iterated integral: integral from 0 to 5 integral from 0 to 4 of  $xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 4] \times [0, 5]$$

Answer: \_\_\_\_\_

5. Evaluate the double integral: integral from 2 to 4 integral from 1 to 3 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [1, 3] \times [2, 4]$$

Answer: \_\_\_\_\_



6. A density function  $\rho = 7 \text{ kg/m}^2$  is uniform over the rectangle  $[0,5] \times [1,4]$ . Find the total mass.

$$\iint_R 7 \, dA, [0, 5] \times [1, 4]$$

Answer: \_\_\_\_\_

7. Evaluate: integral from 0 to 3 integral from 0 to 4 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 4] \times [0, 3]$$

Answer: \_\_\_\_\_

8. Evaluate the iterated integral:  $\int_0^4 \int_0^2 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 2] \times [0, 4]$$

Answer: \_\_\_\_\_

9. Evaluate the double integral: integral from 2 to 4 integral from 1 to 5 of  $8 \, dx \, dy$ .

$$\iint_R 8 \, dA, [1, 5] \times [2, 4]$$

Answer: \_\_\_\_\_

10. A density function  $\rho = 5 \text{ kg/m}^2$  is uniform over the rectangle  $[0,3] \times [0,4]$ . Find the total mass.

$$\iint_R 5 \, dA, [0, 3] \times [0, 4]$$

Answer: \_\_\_\_\_

11. Evaluate: integral from 0 to 2 integral from 0 to 4 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 4] \times [0, 2]$$

Answer: \_\_\_\_\_

12. Evaluate the iterated integral:  $\int_0^3 \int_0^6 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 6] \times [0, 3]$$

Answer: \_\_\_\_\_

13. Evaluate the double integral: integral from 1 to 6 integral from 1 to 5 of  $3 \, dx \, dy$ .

$$\iint_R 3 \, dA, [1, 5] \times [1, 6]$$

Answer: \_\_\_\_\_



14. A density function  $\rho = 3 \text{ kg/m}^2$  is uniform over the rectangle  $[0,3] \times [1,2]$ . Find the total mass.

$$\iint_R 3 \, dA, [0, 3] \times [1, 2]$$

Answer: \_\_\_\_\_

15. Evaluate: integral from 0 to 5 integral from 0 to 2 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 2] \times [0, 5]$$

Answer: \_\_\_\_\_

16. Evaluate the iterated integral:  $\int_0^4 \int_0^2 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 2] \times [0, 4]$$

Answer: \_\_\_\_\_

17. Evaluate the double integral: integral from 0 to 4 integral from 0 to 4 of  $4 \, dx \, dy$ .

$$\iint_R 4 \, dA, [0, 4] \times [0, 4]$$

Answer: \_\_\_\_\_

18. A density function  $\rho = 5 \text{ kg/m}^2$  is uniform over the rectangle  $[0,5] \times [0,3]$ . Find the total mass.

$$\iint_R 5 \, dA, [0, 5] \times [0, 3]$$

Answer: \_\_\_\_\_

19. Evaluate: integral from 0 to 4 integral from 0 to 4 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 4] \times [0, 4]$$

Answer: \_\_\_\_\_

20. Evaluate the iterated integral:  $\int_0^3 \int_0^5 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 5] \times [0, 3]$$

Answer: \_\_\_\_\_

21. Evaluate the double integral: integral from 1 to 5 integral from 0 to 4 of  $3 \, dx \, dy$ .

$$\iint_R 3 \, dA, [0, 4] \times [1, 5]$$

Answer: \_\_\_\_\_



22. A density function  $\rho = 9 \text{ kg/m}^2$  is uniform over the rectangle  $[1,4] \times [1,5]$ . Find the total mass.

$$\iint_R 9 \, dA, [1, 4] \times [1, 5]$$

Answer: \_\_\_\_\_

23. Evaluate: integral from 0 to 2 integral from 0 to 5 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 5] \times [0, 2]$$

Answer: \_\_\_\_\_

24. Evaluate the iterated integral:  $\int_0^4 \int_0^5 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 5] \times [0, 4]$$

Answer: \_\_\_\_\_

25. Evaluate the double integral: integral from 0 to 6 integral from 1 to 6 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [1, 6] \times [0, 6]$$

Answer: \_\_\_\_\_

26. A density function  $\rho = 3 \text{ kg/m}^2$  is uniform over the rectangle  $[0,5] \times [1,5]$ . Find the total mass.

$$\iint_R 3 \, dA, [0, 5] \times [1, 5]$$

Answer: \_\_\_\_\_

27. Evaluate: integral from 0 to 2 integral from 0 to 5 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 5] \times [0, 2]$$

Answer: \_\_\_\_\_

28. Evaluate the iterated integral:  $\int_0^2 \int_0^2 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 2] \times [0, 2]$$

Answer: \_\_\_\_\_

29. Evaluate the double integral: integral from 1 to 4 integral from 1 to 6 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [1, 6] \times [1, 4]$$

Answer: \_\_\_\_\_



30. A density function  $\rho = 7 \text{ kg/m}^2$  is uniform over the rectangle  $[0,4] \times [0,3]$ . Find the total mass.

$$\iint_R 7 \, dA, [0, 4] \times [0, 3]$$

Answer: \_\_\_\_\_

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# MATH230: Double Integrals

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ANSWER KEY & SOLUTIONS

*Topics: Double integrals over rectangles. All answers verified by independent computation.*

## Solutions

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## Double integrals over rectangles

1. Evaluate the double integral: integral from 0 to 4 integral from 0 to 5 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [0, 5] \times [0, 4]$$

→ Inner integral: integral from 0 to 5 of  $2 \, dx = 2 \cdot (5-0) = 2 \cdot 5$ .

→ Outer integral: integral from 0 to 4 of  $2 \cdot 5 \, dy = 2 \cdot 5 \cdot (4-0) = 40$ .

**Answer:**  $= 2 \cdot 5 \cdot 4 = 40$

2. A density function  $\rho = 2 \, \text{kg/m}^2$  is uniform over the rectangle  $[1,3] \times [1,4]$ . Find the total mass.

$$\iint_R 2 \, dA, [1, 3] \times [1, 4]$$

→ Mass = integral integral  $\rho \, dA = 2 \cdot (\text{width}) \cdot (\text{height})$ .

→  $= 2 \cdot (3-1) \cdot (4-1) = 2 \cdot 2 \cdot 3 = 12 \, \text{kg}$ .

**Answer:**  $= 2 \cdot 2 \cdot 3 = 12$

3. Evaluate: integral from 0 to 2 integral from 0 to 5 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 5] \times [0, 2]$$

→ Inner: integral  $\int_0^5 xy \, dx = x^2 y / 2$  from 0 to 5  $= 5^2 y / 2$ .

→ Outer: integral  $\int_0^2 (5^2 y / 2) \, dy = 5^2 / 2 \cdot y^2 / 2$  from 0 to 2  $= 5^2 \cdot 2^2 / 4 = 25$ .

**Answer:**  $= \frac{25}{2} \cdot \frac{4}{2} = 25$

4. Evaluate the iterated integral: integral  $\int_0^5 \int_0^4 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 4] \times [0, 5]$$

→ Integrate with respect to  $x$  first: integral  $\int_0^4 xy \, dx = 4^2 y / 2$ .

→ Then with respect to  $y$ : integral  $\int_0^5 4^2 y / 2 \, dy = 16 \cdot 5^2 / 4 = 100$ .

**Answer:**  $= \frac{16}{2} \cdot \frac{25}{2} = 100$

5. Evaluate the double integral: integral from 2 to 4 integral from 1 to 3 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [1, 3] \times [2, 4]$$

→ Inner integral: integral from 1 to 3 of  $2 \, dx = 2 \cdot (3-1) = 2 \cdot 2$ .

→ Outer integral: integral from 2 to 4 of  $2 \cdot 2 \, dy = 2 \cdot 2 \cdot (4-2) = 8$ .

**Answer:**  $= 2 \cdot 2 \cdot 2 = 8$

6. A density function  $\rho = 7 \, \text{kg/m}^2$  is uniform over the rectangle  $[0,5] \times [1,4]$ . Find the total mass.

$$\iint_R 7 \, dA, [0, 5] \times [1, 4]$$

→ Mass = integral integral  $\rho \, dA = 7 \cdot (\text{width}) \cdot (\text{height})$ .

→  $= 7 \cdot (5-0) \cdot (4-1) = 7 \cdot 5 \cdot 3 = 105 \, \text{kg}$ .

**Answer:**  $= 7 \cdot 5 \cdot 3 = 105$



7. Evaluate: integral from 0 to 3 integral from 0 to 4 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 4] \times [0, 3]$$

→ Inner:  $\text{integral}_0^4 xy \, dx = x^2 y/2$  from 0 to 4 =  $4^2 y/2$ .

→ Outer:  $\text{integral}_0^3 (4^2 y/2) \, dy = 4^2/2 \cdot y^2/2$  from 0 to 3 =  $4^2 \cdot 3^2 / 4 = 36$ .

**Answer:**  $= \frac{16}{2} \cdot \frac{9}{2} = 36$

8. Evaluate the iterated integral:  $\text{integral}_0^4 \text{integral}_0^2 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 2] \times [0, 4]$$

→ Integrate with respect to  $x$  first:  $\text{integral}_0^2 xy \, dx = 2^2 y/2$ .

→ Then with respect to  $y$ :  $\text{integral}_0^4 2^2 y/2 \, dy = 416/4 = 16$ .

**Answer:**  $= \frac{4}{2} \cdot \frac{16}{2} = 16$

9. Evaluate the double integral: integral from 2 to 4 integral from 1 to 5 of  $8 \, dx \, dy$ .

$$\iint_R 8 \, dA, [1, 5] \times [2, 4]$$

→ Inner integral: integral from 1 to 5 of  $8 \, dx = 8 \cdot (5-1) = 8 \cdot 4$ .

→ Outer integral: integral from 2 to 4 of  $8 \cdot 4 \, dy = 8 \cdot 4 \cdot (4-2) = 64$ .

**Answer:**  $= 8 \cdot 4 \cdot 2 = 64$

10. A density function  $\rho = 5 \, \text{kg/m}^2$  is uniform over the rectangle  $[0,3] \times [0,4]$ . Find the total mass.

$$\iint_R 5 \, dA, [0, 3] \times [0, 4]$$

→ Mass = integral integral  $\rho \, dA = 5 \cdot (\text{width}) \cdot (\text{height})$ .

→  $= 5 \cdot (3-0) \cdot (4-0) = 5 \cdot 3 \cdot 4 = 60 \, \text{kg}$ .

**Answer:**  $= 5 \cdot 3 \cdot 4 = 60$

11. Evaluate: integral from 0 to 2 integral from 0 to 4 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 4] \times [0, 2]$$

→ Inner:  $\text{integral}_0^4 xy \, dx = x^2 y/2$  from 0 to 4 =  $4^2 y/2$ .

→ Outer:  $\text{integral}_0^2 (4^2 y/2) \, dy = 4^2/2 \cdot y^2/2$  from 0 to 2 =  $4^2 \cdot 2^2 / 4 = 16$ .

**Answer:**  $= \frac{16}{2} \cdot \frac{4}{2} = 16$





12. Evaluate the iterated integral:  $\int_0^3 \int_0^6 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 6] \times [0, 3]$$

→ Integrate with respect to  $x$  first:  $\int_0^6 xy \, dx = 6^2 y/2$ .

→ Then with respect to  $y$ :  $\int_0^3 6^2 y/2 \, dy = 369/4 = 81$ .

**Answer:**  $= \frac{36}{2} \cdot \frac{9}{2} = 81$

13. Evaluate the double integral: integral from 1 to 6 integral from 1 to 5 of  $3 \, dx \, dy$ .

$$\iint_R 3 \, dA, [1, 5] \times [1, 6]$$

→ Inner integral: integral from 1 to 5 of  $3 \, dx = 3 \cdot (5-1) = 3 \cdot 4$ .

→ Outer integral: integral from 1 to 6 of  $3 \cdot 4 \, dy = 3 \cdot 4 \cdot (6-1) = 60$ .

**Answer:**  $= 3 \cdot 4 \cdot 5 = 60$

14. A density function  $\rho = 3 \, \text{kg/m}^2$  is uniform over the rectangle  $[0, 3] \times [1, 2]$ . Find the total mass.

$$\iint_R 3 \, dA, [0, 3] \times [1, 2]$$

→ Mass = integral integral  $\rho \, dA = 3 \cdot (\text{width}) \cdot (\text{height})$ .

→  $= 3 \cdot (3-0) \cdot (2-1) = 3 \cdot 3 \cdot 1 = 9 \, \text{kg}$ .

**Answer:**  $= 3 \cdot 3 \cdot 1 = 9$

15. Evaluate: integral from 0 to 5 integral from 0 to 2 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 2] \times [0, 5]$$

→ Inner:  $\int_0^2 xy \, dx = x^2 y/2$  from 0 to 2  $= 2^2 y/2$ .

→ Outer:  $\int_0^5 (2^2 y/2) \, dy = 2^2/2 \cdot y^2/2$  from 0 to 5  $= 2^2 \cdot 5^2 / 4 = 25$ .

**Answer:**  $= \frac{4}{2} \cdot \frac{25}{2} = 25$

16. Evaluate the iterated integral:  $\int_0^4 \int_0^2 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 2] \times [0, 4]$$

→ Integrate with respect to  $x$  first:  $\int_0^2 xy \, dx = 2^2 y/2$ .

→ Then with respect to  $y$ :  $\int_0^4 2^2 y/2 \, dy = 416/4 = 16$ .

**Answer:**  $= \frac{4}{2} \cdot \frac{16}{2} = 16$

17. Evaluate the double integral: integral from 0 to 4 integral from 0 to 4 of  $4 \, dx \, dy$ .

$$\iint_R 4 \, dA, [0, 4] \times [0, 4]$$

→ Inner integral: integral from 0 to 4 of  $4 \, dx = 4 \cdot (4-0) = 4 \cdot 4$ .

→ Outer integral: integral from 0 to 4 of  $4 \cdot 4 \, dy = 4 \cdot 4 \cdot (4-0) = 64$ .

**Answer:**  $= 4 \cdot 4 \cdot 4 = 64$



18. A density function  $\rho = 5 \text{ kg/m}^2$  is uniform over the rectangle  $[0,5] \times [0,3]$ . Find the total mass.

$$\iint_R 5 \, dA, [0, 5] \times [0, 3]$$

→ Mass = integral integral  $\rho \, dA = 5 \cdot (\text{width}) \cdot (\text{height})$ .

$$\rightarrow = 5 \cdot (5-0) \cdot (3-0) = 5 \cdot 5 \cdot 3 = 75 \text{ kg.}$$

**Answer:**  $= 5 \cdot 5 \cdot 3 = 75$

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19. Evaluate: integral from 0 to 4 integral from 0 to 4 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 4] \times [0, 4]$$

→ Inner: integral  $_0^4 xy \, dx = x^2 y/2$  from 0 to 4 =  $4^2 y/2$ .

→ Outer: integral  $_0^4 (4^2 y/2) \, dy = 4^2/2 \cdot y^2/2$  from 0 to 4 =  $4^2 \cdot 4^2 / 4 = 64$ .

**Answer:**  $= \frac{16}{2} \cdot \frac{16}{2} = 64$

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20. Evaluate the iterated integral: integral  $_0^3$  integral  $_0^5 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 5] \times [0, 3]$$

→ Integrate with respect to  $x$  first: integral  $_0^5 xy \, dx = 5^2 y/2$ .

→ Then with respect to  $y$ : integral  $_0^3 5^2 y/2 \, dy = 259/4 = 225/4$ .

**Answer:**  $= \frac{25}{2} \cdot \frac{9}{2} = \frac{225}{4}$

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21. Evaluate the double integral: integral from 1 to 5 integral from 0 to 4 of  $3 \, dx \, dy$ .

$$\iint_R 3 \, dA, [0, 4] \times [1, 5]$$

→ Inner integral: integral from 0 to 4 of  $3 \, dx = 3 \cdot (4-0) = 3 \cdot 4$ .

→ Outer integral: integral from 1 to 5 of  $3 \cdot 4 \, dy = 3 \cdot 4 \cdot (5-1) = 48$ .

**Answer:**  $= 3 \cdot 4 \cdot 4 = 48$

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22. A density function  $\rho = 9 \text{ kg/m}^2$  is uniform over the rectangle  $[1,4] \times [1,5]$ . Find the total mass.

$$\iint_R 9 \, dA, [1, 4] \times [1, 5]$$

→ Mass = integral integral  $\rho \, dA = 9 \cdot (\text{width}) \cdot (\text{height})$ .

$$\rightarrow = 9 \cdot (4-1) \cdot (5-1) = 9 \cdot 3 \cdot 4 = 108 \text{ kg.}$$

**Answer:**  $= 9 \cdot 3 \cdot 4 = 108$

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23. Evaluate: integral from 0 to 2 integral from 0 to 5 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 5] \times [0, 2]$$

→ Inner: integral  $_0^5 xy \, dx = x^2 y/2$  from 0 to 5 =  $5^2 y/2$ .

→ Outer: integral  $_0^2 (5^2 y/2) \, dy = 5^2/2 \cdot y^2/2$  from 0 to 2 =  $5^2 \cdot 2^2 / 4 = 25$ .

**Answer:**  $= \frac{25}{2} \cdot \frac{4}{2} = 25$

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24. Evaluate the iterated integral:  $\int_0^4 \int_0^5 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 5] \times [0, 4]$$

→ Integrate with respect to  $x$  first:  $\int_0^5 xy \, dx = 5^2 y/2$ .

→ Then with respect to  $y$ :  $\int_0^4 5^2 y/2 \, dy = 2516/4 = 100$ .

**Answer:**  $= \frac{25}{2} \cdot \frac{16}{2} = 100$

25. Evaluate the double integral: integral from 0 to 6 integral from 1 to 6 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [1, 6] \times [0, 6]$$

→ Inner integral: integral from 1 to 6 of  $2 \, dx = 2 \cdot (6-1) = 2 \cdot 5$ .

→ Outer integral: integral from 0 to 6 of  $2 \cdot 5 \, dy = 2 \cdot 5 \cdot (6-0) = 60$ .

**Answer:**  $= 2 \cdot 5 \cdot 6 = 60$

26. A density function  $\rho = 3 \, \text{kg/m}^2$  is uniform over the rectangle  $[0,5] \times [1,5]$ . Find the total mass.

$$\iint_R 3 \, dA, [0, 5] \times [1, 5]$$

→ Mass = integral integral  $\rho \, dA = 3 \cdot (\text{width}) \cdot (\text{height})$ .

→  $= 3 \cdot (5-0) \cdot (5-1) = 3 \cdot 5 \cdot 4 = 60 \, \text{kg}$ .

**Answer:**  $= 3 \cdot 5 \cdot 4 = 60$

27. Evaluate: integral from 0 to 2 integral from 0 to 5 of  $xy \, dx \, dy$ .

$$\iint_R xy \, dA, [0, 5] \times [0, 2]$$

→ Inner:  $\int_0^5 xy \, dx = x^2 y/2$  from 0 to 5 =  $5^2 y/2$ .

→ Outer:  $\int_0^2 (5^2 y/2) \, dy = 5^2/2 \cdot y^2/2$  from 0 to 2 =  $5^2 \cdot 2^2 / 4 = 25$ .

**Answer:**  $= \frac{25}{2} \cdot \frac{4}{2} = 25$

28. Evaluate the iterated integral:  $\int_0^2 \int_0^2 xy \, dx \, dy$ . Use Fubini's theorem to evaluate as an iterated integral.

$$\iint_R xy \, dA, [0, 2] \times [0, 2]$$

→ Integrate with respect to  $x$  first:  $\int_0^2 xy \, dx = 2^2 y/2$ .

→ Then with respect to  $y$ :  $\int_0^2 2^2 y/2 \, dy = 44/4 = 4$ .

**Answer:**  $= \frac{4}{2} \cdot \frac{4}{2} = 4$

29. Evaluate the double integral: integral from 1 to 4 integral from 1 to 6 of  $2 \, dx \, dy$ .

$$\iint_R 2 \, dA, [1, 6] \times [1, 4]$$

→ Inner integral: integral from 1 to 6 of  $2 \, dx = 2 \cdot (6-1) = 2 \cdot 5$ .

→ Outer integral: integral from 1 to 4 of  $2 \cdot 5 \, dy = 2 \cdot 5 \cdot (4-1) = 30$ .

**Answer:**  $= 2 \cdot 5 \cdot 3 = 30$



30. A density function  $\rho = 7 \text{ kg/m}^2$  is uniform over the rectangle  $[0,4] \times [0,3]$ . Find the total mass.

$$\iint_R 7 \, dA, [0, 4] \times [0, 3]$$

→ Mass = integral integral  $\rho \, dA = 7 \cdot (\text{width}) \cdot (\text{height})$ .

→  $= 7 \cdot (4-0) \cdot (3-0) = 7 \cdot 4 \cdot 3 = 84 \text{ kg}$ .

**Answer:**  $= 7 \cdot 4 \cdot 3 = 84$

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